



LUX INDUSTRIES

Overview of IP Licensing and Liquidity.io Fraud

Prepared for: Cyrus Pahlavi, President & CEO

Hunter Dupont, Managing Partner — Strategic Transactions

Alan Donohue, General Counsel

Prepared by: Woo Bin — CTO, Lux Industries Inc. / VP Engineering, Satschel Inc.

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MEMORANDUM

TO: Cyrus Pahlavi, President & Chief Executive Officer, Lux Industries Inc.
Hunter Dupont, Managing Partner — Strategic Transactions (deal lead)
Alan Donohue, General Counsel (legal lead)

CC: Board of Directors, Lux Industries Inc.

FROM: Woo Bin — Chief Technology Officer, Lux Industries Inc.; concurrently Vice President of Engineering, Satschel, Inc. d/b/a Liquidity.io

DATE: 25 May 2026

RE: Overview of IP Licensing and Liquidity.io Fraud — Misappropriation of Lux foundational technology by Satschel, Inc. and Liquidity.io LLC; legal options, damages analysis, and proposed settlement framework

PRIVILEGE: Confidential. Attorney-client privileged. Attorney work product. Prepared in anticipation of litigation and as a board strategy document.

Author's disclosure of interest: I serve concurrently as Chief Technology Officer of Lux Industries Inc. and Vice President of Engineering at Satschel, Inc. (d/b/a Liquidity.io). This memorandum is written from the Lux chair, in service of the Lux board's decision-making. My Satschel role gives direct operational visibility into the matters described; I have set them out as observed, with the understanding that any Satschel-side response or counter-position will need to be developed by Satschel's own counsel and that Lux's outside litigation counsel will need to develop independent record support for any allegations herein before filing.

I. Executive Summary

Satschel, Inc. and its operating affiliate Liquidity.io LLC (collectively, “Satschel” or “Liquidity”) have commercially deployed material portions of Lux Industries Inc.'s patent-protected and source-available proprietary technology since at least 2014, in production, on three live blockchain networks (Liquid Devnet ChainID 8675312; Liquid Testnet ChainID 8675310; Liquid Mainnet ChainID 8675309), and in support of a registered ATS/BD/TA business that recently raised approximately \$1.1 billion in capital at a post-money valuation in that range — in each case without executing the commercial license that the governing Lux Ecosystem License v1.2 (“LEL”) and the Lux Research License with Patent Reservation v1.0 (“LRL-PR”) require.

Lux's investment in the misappropriated technology base exceeds \$20 million in direct development cost recorded against Lux books from 2014 to date, plus a multi-year contribution from senior engineering personnel including the named author of this memorandum, the Lux Chief Cryptographer, and others.

This memorandum sets out the legal record, identifies the causes of action available to Lux against Satschel and its officers, quantifies damages at small / medium / large scenarios, projects the forward enterprise value of the equity component at user-scale breakpoints of 100K, 1M, 10M, and 100M users, analyzes the “wait vs. settle” time-value question, and offers a settlement triangulation matrix the deal lead (Mr. Dupont) and General Counsel (Mr. Donohue) can use to negotiate.

This is not a single-proposal document. It presents the data, the damages range, the partnership-upside range, and the negotiating floors — the Lux deal team can triangulate to a specific outcome. The final approval authority on any executed Settlement and Cross- License Agreement (§XII) rests

with the President & CEO, Mr. Pahlavi.

II. Parties and Dramatis Personae

Lux Industries Inc. — Decision Chain

- **Cyrus Pahlavi** — President & Chief Executive Officer. Final approval authority on any settlement, license, or litigation decision.
- **Hunter Dupont** — Largest investor in Lux Industries Inc.; Managing Partner, Strategic Transactions. Leads deal discussions and commercial negotiations with Satschel.
- **Alan Donohue** — General Counsel. Leads legal strategy and definitive-document drafting; provides legal support to Mr. Dupont throughout the negotiation.
- **Woo Bin** — Chief Technology Officer. Author of this memorandum. Holds concurrent role as Vice President of Engineering at Satschel, Inc.; operational visibility into Satschel's use of Lux IP.
- **Zach Kelling** — Founder, Architect, and Chief Cryptographer of Lux. Minority shareholder. **See §X for particularized treatment.** Not a director on either side. Not the current CTO of either entity.

Satschel, Inc. d/b/a Liquidity.io — Counterparty Officer Slate

- **Eric Choi** — Chief Executive Officer. Top operational authority and the officer with ultimate responsibility for Satschel's representations to investors and counterparties regarding IP ownership and licensing.
- **Austin Tromblye** — Chief Strategy Officer; previously Acting Chief Executive Officer. Blockchain / AI engineer. Built a **prior** (not current) ATS infrastructure at a prior employer (Franklin Templeton); served as an eyewitness in the Franklin Templeton litigation arising from that engagement. **Counsel should review the Franklin Templeton litigation record** for (i) pattern-of- conduct evidence relevant to the present matter, (ii) any limits on or adverse characterizations of his prior testimony that could be impeachable, and (iii) the technical scope and quality of the ATS he built there. Material context: prior to Mr. Kelling's arrival as active Chief Technology Officer at Satschel (see *infra* §IV(G)), the Satschel ATS as built under Mr. Tromblye's technical direction **lacked even basic time-price priority order matching, the foundational principle of any functioning alternative trading system.** The technical state of the pre-Kelling Satschel platform is addressed in §IV(G) and bears directly on Mr. Tromblye's credibility as a technology witness in any subsequent proceeding.
- **Coleman Church** — Chief Executive Officer, Liquidity Broker-Dealer. Responsible for BD-side compliance posture and FINRA representations.
- **Jayant Sonsurkar** — Chief Security Officer. Long- standing maintainer / admin on the `luxfi/*` GitHub organization. The CSO role at any regulated digital-asset venue carries the duty to oversee software-supply-chain integrity, including open-source license review and third-party-IP compliance. As maintainer on every Lux repository Satschel later deployed, Mr. Sonsurkar had on-platform visibility into the LEL, LRL-PR, BSD-3, and other LICENSE files as they were

created, amended, and committed. The license tier of each Lux module was directly observable in commit history under his admin access.

- **Zach Kelling** — Founder of Satschel; Architect; Chief Cryptographer. **Not the current CEO or CTO of Satschel; previously held those titles in earlier corporate periods. Minority shareholder of Lux Industries Inc. and zero-holder of any Satschel / Liquidity.io equity.** Effectively, on the Satschel side, Mr. Kelling has been operationally displaced from the executive roles he previously held and now occupies an architect / chief- cryptographer position with limited day-to-day decision authority over the commercial direction. **See §X.**

Counterparty Entity Structure

For purposes of this memorandum, “Satschel” and “Liquidity” refer collectively to Satschel, Inc. (a Delaware corporation) and Liquidity.io LLC (its operating affiliate), unless context requires otherwise. The corporate structure should be confirmed during outside- counsel diligence.

III. Jurisdiction, Venue, and Applicable Law

Federal causes of action:

- **Defend Trade Secrets Act (DTSA)** — 18 U.S.C. §§1836–1839. Federal question jurisdiction under 28 U.S.C. §1331; nationwide service of process under DTSA §1836(c).
- **Copyright Act** — 17 U.S.C. §§101 *et seq.* Exclusive federal jurisdiction under 28 U.S.C. §1338(a).
- **Patent Act** — 35 U.S.C. §§271, 281, 284, 285. Exclusive federal jurisdiction; Federal Circuit on appeal.
- **Lanham Act** — 15 U.S.C. §1125(a), false advertising / false designation of origin.
- **Securities Exchange Act of 1934** — 15 U.S.C. §78j(b); Rule 10b-5, 17 C.F.R. §240.10b-5. Federal question jurisdiction.

State / common-law causes of action:

- **Delaware Breach of Fiduciary Duty** — against Satschel officers individually. Court of Chancery; see *Aronson v. Lewis*, 473 A.2d 805 (Del. 1984); *Smith v. Van Gorkom*, 488 A.2d 858 (Del. 1985).
- **Aiding and Abetting Breach of Fiduciary Duty** — *Malpiede v. Townson*, 780 A.2d 1075 (Del. 2001).
- **Common-Law Fraud** — Delaware and California (depending on contracting venue).
- **Unjust Enrichment** — Delaware / California.
- **Conversion** — Delaware / California.
- **Tortious Interference with Prospective Economic Advantage** — California (Liquidity’s principal place of business if San Francisco-based).
- **Civil Conspiracy** — Delaware / California.

Forum recommendation: Federal District Court for the Northern District of California (likely Liquidity’s principal place of business and the locus of the trade secret misappropriation), with related Delaware Chancery action for the fiduciary-duty + corporate-governance claims. Diversity of citizenship and federal-question jurisdiction both available. Personal jurisdiction over individual Satschel officers established by their commercial activities directed at Lux (a Delaware corporation) and / or their California residence.

Litigation hold — Lux should issue a formal litigation hold notice to Satschel within 30 days of the SCLA opening offer, addressed to the corporate entity and named individually to Messrs. Choi, Tromblye, Church, and Sonsurkar. Failure of Satschel to preserve records following such notice supports an adverse-inference instruction at trial.

IV. Statement of Facts

The following numbered allegations are stated on information and belief, based on the personal knowledge of the undersigned in his dual operational role, supplemented by Lux’s internal records. Each will need to be verified, supplemented, or amended through outside- counsel-supervised diligence prior to any filing.

A. Background of Lux Industries Inc. and Its IP (2014–2026)

1. Lux Industries Inc. (the predecessor entity Lux Partners Limited, with continuous corporate identity since 2014) has continuously developed, since 2014, a blockchain and cryptographic technology stack including (without limitation) consensus algorithms, post-quantum cryptographic primitives, threshold and multi-party-computation protocols, fully-homomorphic-encryption (FHE) precompile architecture, virtual-machine substrates, EVM derivatives, decentralized-exchange matching engines, and cross-chain messaging protocols.
2. Lux’s continuous direct development investment in this technology stack, recorded against Lux’s books from 2014 to the date of this memorandum, exceeds \$20 million in direct cost. Fully-loaded personnel cost over the same period — senior engineering capacity not allocated as project cost — substantially exceeds the direct- cost figure.
3. Lux’s chronological development milestones include:
 - **2014 – 2017.** Foundational consensus research; early Quasar consensus prototypes; first cryptographic primitives library; `secp256k1`, `Ed25519`, `BLS12-381` reference implementations; HD wallet (BIP-32 / BIP-39) contributions.
 - **2018 – 2020.** Active development of the decentralized exchange (“DEX”) matching engine, including the planet-scale order-book matching architecture with 1ms finality. Initial EVM fork from Ava Labs subnet-evm under LGPL inheritance. Threshold-MPC research (precursors to CGGMP21 and FROST integrations).
 - **2020 – 2023.** Quasar consensus stabilization; Pulsar and Magnetar Apache-2.0-licensed threshold-PQ primitives; FHE precompile prototypes; precompile registry architecture.
 - **2024.** Ringtail lattice-signature scheme; post-quantum standardization push (FIPS 203 / 204 / 205 verifier precompiles covering ML-KEM, ML-DSA, SLH-DSA); P3Q (Post-Quantum Proof Verifier) protocol; LEL v1.2 and LRL-PR v1.0 formalized; Warp messaging protocol

stabilization; FHE-based dark-pool matching; multi-chain architecture (Lux Z-Chain, Q-Chain, X-Chain).

- **2025.** LX Suite DEX precompile family (Pool / Oracle / Router / Hooks at LP-9010 onward), released following Uniswap v4’s hooks model with native on-chain implementation across Lux precompile addresses; Corona Ring-LWE threshold scheme (successor to Ringtail); LP-4200 precompile address standardization across the `luxfi/precompile/*` registry; multi-arch GPU acceleration under the closed `~/work/luxcpp/*` tree (CUDA, Metal, WebGPU); AI inference precompile (LP-9080 / Compute marketplace).
- **2026.** Brand-neutral cleanup across Lux upstream; relicensing of Pulsar / Magnetar under Apache-2.0 for ecosystem-wide adoption; ongoing institutional partnership development.

4. Lux holds (or has filed for) patents and patent applications on, without limitation: the Quasar BFT consensus algorithm; the sampler-bias correction mechanism; the Ringtail and Corona lattice threshold schemes; the Unified VM Execution Interface with Pluggable Backends; the Post-Quantum State Verification with Verkle Proofs protocol; the FHE precompile architecture; the GPU- accelerated DEX matching engine; and AI-optimized bytecode execution with JIT compilation.

B. The Lux Licensing Tier Structure

5. Lux publishes its software under a three-tier licensing strategy formalized in the internal LICENSING-POLICY (the “Licensing Policy”):

6. Tier 1 (Commodity / BSD-3-Clause). Reference clients, SDKs, common utilities, and other widely-useful but non- defensible code. Commercial use is free under standard BSD-3 attribution. Examples: `luxfi/node`, `luxfi/sdk`, `luxfi/cli`, `luxfi/api`, `luxfi/codecs`, `luxfi/database`, `luxfi/p2p`, and approximately twenty- five (25) other modules. Liquidity uses these freely with no Lux action required.

7. Tier 2 (Lux Ecosystem License v1.2, “LEL”). Patent-protected source-available code. Lux publishes source for inspection, research, audit, and Authorized-Network commercial use, but reserves commercial rights outside the Lux ecosystem. The license text at `~/work/lux/consensus/LICENSE` controls. Per LEL §§1 and 3, rights are granted only to “Authorized Networks” — defined to include the Lux Primary Network (NetworkID 1, ChainID 96369), Lux testnets and devnets, and “L1/L2/L3 chains descending from the Lux Primary Network.”

8. Per LEL §4, forks and competing networks are explicitly prohibited. Per LEL §5, all patent rights are reserved; the copyright grant does not include any patent license. Per LEL §8, the license terminates automatically on breach. Modules under LEL include: `luxfi/consensus`, `luxfi/crypto`, `luxfi/mpc`, `luxfi/threshold`, `luxfi/ringtail`, `luxfi/corona`, `luxfi/fhe`, `luxfi/precompile` (and sub-precompiles), `luxfi/captable`, `luxfi/broker`, `luxfi/treasury`, `luxfi/transfer`, `luxfi/compliance`, `luxfi/hsm`, `luxfi/aml`, `luxfi/maker`, `luxfi/warp`, among others.

9. Tier 3 (Lux Research License with Patent Reservation v1.0, “LRL-PR”). Research-only commercial license. Per the LRL-PR text at `~/work/lux/vm/LICENSE`, Section 2 grants only research-use rights; Section 3 expressly prohibits any commercial use without separate written license from Lux; Section 4 expressly reserves all patent claims; Section 5 auto- licenses validator-node operation only on Lux mainnet and testnets (no Descending-Chain carve-out); Section 6 contains anti-patent- filing covenants. Modules under LRL-PR include: `luxfi/vm`, `luxfi/staking`,

luxfi/sampler, and the Go bindings of luxfi/dex prior to that repository’s move to fully proprietary status in 2026.

10. Tier 4 (Closed Proprietary). The ~/work/luxcpp/* tree (CUDA / Metal / WebGPU / FPGA / lattice GPU accelerators, the C++ DEX matching engine, and related code) is private repository code requiring a separately negotiated Private-Moat Addendum and commercial license. Liquidity does not currently use this tier; access in the future requires a separate written agreement.

C. Satschel’s Commercial Use of Lux IP

11. Satschel (through Satschel, Inc. and Liquidity.io LLC) operates a registered Alternative Trading System (“ATS”), broker-dealer (“BD”), and transfer-agent (“TA”) business in the United States. Satschel commercially deploys, in production, on three live blockchain networks — Liquid Devnet (ChainID 8675312), Liquid Testnet (ChainID 8675310), and Liquid Mainnet (ChainID 8675309) — the validator binary lqd, built from the liquidityio/node source tree, which in turn imports and incorporates Lux modules including luxfi/node, luxfi/consensus, luxfi/crypto, luxfi/evm, luxfi/vm, and dozens of others.

12. Satschel further deploys, in production:

- The Liquidity EVM (liquidityio/evm), which incorporates luxfi/evm (LGPL-3.0, with §6 source-offer obligations), luxfi/vm (LRL-PR), and luxfi/precompile (LEL with sub-modules).
- The Liquidity ATS, BD, and TA backend services, which import and use luxfi/broker, luxfi/treasury, luxfi/transfer, luxfi/captable, luxfi/compliance, luxfi/hsm, luxfi/aml, luxfi/maker, luxfi/crypto, luxfi/mpc, luxfi/threshold, and luxfi/fhe — all LEL-licensed.
- Multiple white-label tenant deployments (including MLC, VCC, TZERO, ReTokens, and other Satschel-controlled tenants) operating on the same Lux-licensed substrate.
- 12,796 SecurityTokens deployed on the Liquid Devnet alone, each running on top of Lux pre-compiles for signature verification, threshold operations, FHE-protected dark-pool matching, and on-chain identity / compliance.

13. The audit underlying this memorandum (LIQUIDITY_LICENSE_AUDIT_2026_05_25.md, co-located in the Lux Industries Inc. legal directory) inventories every Lux module directly imported by Satschel and assigns each its operative license tier. Approximately fifteen (15) directly-imported Lux modules carry LEL or LRL-PR tier licenses requiring an executed commercial license for Satschel’s mode of use.

14. No commercial license has ever been executed between Lux Industries Inc. and Satschel. No side letter, no oral agreement reduced to writing, no Board resolution authorizing the use exists in Lux’s records to date. Outside counsel should verify the non-existence of any such instrument through document discovery during the negotiation or litigation phase.

15. The Liquid blockchain networks operated by Satschel do not currently activate the Lux Warp messaging precompile (canonical address 0x020000000000000000000000000000000005). Specifically, the warpConfig block is absent from the genesis JSON of each Liquid EVM chain. Under the LEL Authorized- Network definition, a chain that does not maintain Lux consensus and Warp interop does not qualify for the LEL ecosystem-permissive grant. **Therefore, even crediting the most generous reading of LEL §1, the Liquid chains do not currently qualify as**

“L1/L2/L3 chains descending from the Lux Primary Network” as a matter of on-chain fact.

D. The Satschel Officer Team and Chain of Knowledge

16. At all relevant times, Mr. Eric Choi served (or now serves) as Chief Executive Officer of Satschel and Liquidity.io. The CEO role at any regulated digital-asset venue carries ultimate operational responsibility for the licensing and intellectual-property compliance of every technology asset used in production. Mr. Choi’s signature appears on the subscription documents for the \$1.1 billion capital raise; representations made therein regarding Satschel’s technology ownership and licensing posture sound in his name.

17. At all relevant times, Mr. Austin Tromblye served as Chief Strategy Officer of Satschel / Liquidity.io, and prior to that served as Acting Chief Executive Officer. Mr. Tromblye is a blockchain and AI engineer by background; he previously built ATS infrastructure for Franklin Templeton and served as an eyewitness in the litigation arising from that engagement. **Counsel should review the Franklin Templeton litigation record** — both for any pattern-of-conduct evidence relevant here, and for material that may inform Mr. Tromblye’s likely posture as a witness in any Lux v. Satschel proceeding (including any limits on his prior testimony that may be impeachable).

18. At all relevant times, Mr. Coleman Church served as Chief Executive Officer of Satschel’s broker-dealer affiliate. As BD CEO, Mr. Church holds FINRA-regulated responsibility for the BD’s technology and operational integrity. His representations on FINRA filings and BD compliance attestations are at risk if those filings do not accurately disclose the Lux licensing status of underlying BD technology.

19. At all relevant times material to this matter, Mr. Jayant Sonsurkar has served as Chief Security Officer of Satschel and has held maintainer / admin access on the GitHub organization `luxfi/*`. The CSO role at any digital-asset venue, and especially at one carrying ATS/BD/TA registrations, includes (without limitation):

- Oversight of software-supply-chain integrity, including third-party-IP review and license compliance.
- Identification of LEL, LRL-PR, GPL, LGPL, and other restricted licenses in deployed dependencies.
- Escalation of licensing mismatches to the CEO, BD CEO, and General Counsel.
- Implementation of remediation when a mismatch is identified.

20. Mr. Sonsurkar’s GitHub admin access to `luxfi/*` gave him direct, ongoing, contemporaneous visibility into:

- Every Lux LICENSE file as it was authored, amended, and committed (including the LEL v1.2 December 2025 finalization, the LRL-PR v1.0 December 2025 finalization, and the relicensing of specific modules over time).
- The complete commit history of every Lux module, including copyright headers, SPDX identifiers, and patent-reservation notices.
- The identity of every Lux author and the chronological record of each contribution.

- Code review and pull-request discussions, including any discussions of license tier and commercial-use posture.

21. On information and belief, Mr. Sonsurkar did not, at any time during the relevant period, raise the LEL / LRL-PR licensing mismatch internally at Satschel; did not request that Satschel execute a commercial license with Lux; did not include the unlicensed-IP status in any FINRA / SEC compliance review; and did not flag the issue during the \$1.1 billion capital-raise diligence process. This omission is the central evidentiary fact supporting the breach-of-fiduciary-duty count (Count IV) and the aiding-and-abetting count (Count V), and tends to support the Path B factual characterization described in §V (intentional misappropriation rather than mere negligence).

22. On information and belief, Messrs. Choi, Tromblye, and Church each had operational and / or strategic visibility into Satschel's technology stack and the licensing posture of its underlying components. The factual record will need to develop through outside-counsel-supervised discovery the precise scope of each officer's actual knowledge.

E. The Position of Mr. Zach Kelling

23. Mr. Zach Kelling is the Founder of both Lux Industries Inc. and Satschel, Inc. He currently serves as Architect and Chief Cryptographer of Satschel. **He is not currently the Chief Executive Officer or Chief Technology Officer of Satschel**, having previously held those titles in earlier corporate periods (which this memorandum acknowledges expressly to avoid any inference of current operational authority on his part).

24. Mr. Kelling is a **minority shareholder** of Lux Industries Inc., and a **zero-holder** of any Satschel / Liquidity.io equity. As such, Mr. Kelling does not stand to gain financially from Satschel's commercial outcome and has not been positioned to direct Satschel's commercial decisions, licensing posture, or representations to investors.

25. The factual record supports a finding that Mr. Kelling has been operationally displaced from Satschel's commercial direction and has not been a principal actor in the conduct giving rise to the claims in this memorandum. The relevant Satschel officers — Messrs. Choi, Tromblye, Church, and Sonsurkar — have operated independently of Mr. Kelling in the matters at issue. Lux's litigation strategy (§ XII) should reflect Mr. Kelling's distinct position.

F. Pre-Kelling Technical State and the ACH Exploit (Material Context)

28. The factual record establishes that, in the period immediately preceding Mr. Zach Kelling's arrival as active Chief Technology Officer of Satschel, the Satschel ATS platform was materially non-functional as a regulated alternative trading system. Specifically, on information and belief:

- The Satschel ATS **did not implement working time-price priority order matching** — the foundational ordering principle of every regulated US alternative trading system, required by SEC Reg ATS for any venue that purports to operate as a securities trading facility.
- No functioning ATS pipeline existed end-to-end. Critical components, including order acceptance, matching engine logic, settlement instructions, and post-trade reporting, were either non-functional, partially built, or built in a manner that did not satisfy basic regulatory or commercial requirements.
- The platform was operationally vulnerable to elementary security exploits. **One day prior to Mr. Kelling's arrival as active CTO, the Satschel platform suffered an ACH-funding**

exploit that resulted in the theft of approximately \$40,000 via a trivial attack vector that would have been remediated by even baseline security review. The exploit underscores both the security posture of the platform under the pre-Kelling officer team and the absence of competent CSO oversight (Mr. Sonsurkar's role) during that period.

29. Upon his arrival as active CTO, Mr. Kelling and his team (substantially overlapping with the Lux Industries Inc. engineering team in personnel and methodology) rebuilt the Satschel ATS, BD, and TA technology stack from the ground up, importing the Lux modules described in §IV(C) as the foundation. **In substantial part, the technology that subsequently supported Satschel's \$1.1 billion capital raise was the Lux IP that Mr. Kelling brought across, integrated with the regulatory wrapper.**

30. The technical leadership transition from the pre- Kelling officer team to the Kelling era is the inflection point that made Satschel a fundable enterprise. The factual proposition that Satschel's enterprise value, as represented to investors in the \$1.1B raise, is materially attributable to Lux-origin technology contributed by Mr. Kelling (and the broader Lux team) is supported by the contemporaneous technical record.

31. This material context bears on (a) the damages attribution analysis in §VI (the share of \$1.1B-valuation enterprise value attributable to Lux contribution); (b) the credibility of Mr. Tromblye as a technology witness in any subsequent proceeding, given the documented prior technical state; (c) the breach-of-fiduciary-duty count (§V, Count IV) and the absence of CSO oversight evidenced by the ACH exploit; and (d) the protective treatment of Mr. Kelling described in §XII (his role was that of remediator and rescuer, not contributor to the underlying misappropriation).

G. The \$1.1 Billion Capital Raise

26. In or around the most recent capital raise, Satschel obtained approximately \$1.1 billion in investor capital at a post- money valuation in that range. The subscription agreements, private placement memorandum ("PPM"), and side letters from that raise will require outside-counsel review for any representations regarding:

- Satschel's ownership of the underlying technology stack.
- Licensing status of all third-party intellectual property.
- Related-party transactions including any Lux-Satschel arrangements.
- The CSO's compliance attestations.
- The BD CEO's compliance attestations.
- Technology disclosures in the risk factors section.

27. To the extent the raise documents represent (or omit material facts regarding) Satschel's ownership or licensed access to Lux IP, those representations and omissions are independently actionable under Section 10(b) of the Securities Exchange Act, 15 U.S.C. §78j(b), and Rule 10b-5, 17 C.F.R. §240.10b-5. See Count IX, *infra*.

V. Causes of Action Available to Lux Industries Inc.

The following causes of action are available based on the factual record summarized in §IV. Each is stated with its statutory or common-law basis, elements, factual support, and applicable damages framework. Outside counsel should refine the pleading drafts in light of jurisdiction-specific pleading requirements and the record developed in pre-suit discovery.

Count I — Misappropriation of Trade Secrets (Defend Trade Secrets Act, 18 U.S.C. §1836)

Elements. Plaintiff must show: (i) ownership of a trade secret; (ii) related to a product or service used in interstate commerce; (iii) that defendant misappropriated; (iv) by improper means or by acquiring with knowledge of improper acquisition.

Factual support. Lux’s unpublished optimizations, integration know-how, architectural documents, deployment recipes, MPC operational procedures, sampler-bias remediation methods, FHE operating envelopes, and pre-publication code in private repositories all qualify as trade secrets. The Liquid chain bootstrap, the MPC custody integration, and the FHE dark-pool matching configurations exposed to Satschel through Mr. Sonsurkar’s repository access constitute interstate use; Mr. Sonsurkar’s continuing access without an executed license, coupled with Satschel’s failure to flag the licensing mismatch despite his duty to do so, supports misappropriation by improper means.

Remedies. Injunctive relief, 18 U.S.C. §1836(b)(3)(A); actual damages and unjust enrichment, §1836(b)(3)(B); reasonable royalty in lieu, *id.*; exemplary damages up to 2× for willful and malicious misappropriation, §1836(b)(3)(C); attorneys’ fees, §1836(b)(3)(D); seizure order if record warrants, §1836(b)(2).

Count II — Copyright Infringement (17 U.S.C. §501)

Elements. Plaintiff must show: (i) ownership of valid copyright; (ii) copying of constituent elements of the work that are original.

Factual support. Lux owns copyright in every source file under the `luxfi/*` GitHub organization. Each Lux author’s contribution vests copyright in Lux under standard assignment language; copyright notices are present in source headers. Satschel has copied — by direct import in `go.mod`, by container- image embedding, by binary distribution to validator nodes, and by on-chain deployment of derivative SecurityTokens — substantially all constituent elements of the LEL and LRL-PR-licensed Lux source.

Remedies. Actual damages and infringer’s profits, 17 U.S.C. §504(b); or statutory damages up to \$150,000 per work willfully infringed, §504(c)(2); injunction, §502; impoundment and destruction of infringing copies, §503; attorneys’ fees, §505; costs, §505.

Count III — Patent Infringement (35 U.S.C. §271)

Elements. Plaintiff must show: (i) ownership of valid, enforceable patent claim; (ii) infringement by making, using, selling, offering for sale, or importing into the United States without authority.

Factual support. Lux holds or has filed for patents covering the inventions enumerated in §IV(A)(4), *supra*. Satschel’s operation of validator nodes implementing Quasar consensus, its deployment of Ringtail / Corona threshold signatures, its use of the sampler-bias correction mecha-

nism, its operation of the Liquid EVM substrate implementing the Unified VM Execution Interface, its FHE precompile activations, and its DEX matching engine each implicate one or more patent claims. The LRL-PR text §4 expressly reserves the patent rights, so the copyright license that Satschel might otherwise argue (and which it cannot validly argue) does not include any patent grant.

Remedies. Reasonable royalty (the statutory floor), 35 U.S.C. §284; lost profits where established; injunctive relief, §283; treble damages for willful infringement, §284; attorneys' fees in exceptional cases, §285; pre- and post-judgment interest.

Count IV — Breach of Fiduciary Duty (Delaware Law, against Jayant Sonsurkar individually)

Elements. As a corporate officer, defendant owes the corporation duties of care and loyalty. Breach occurs upon failure to exercise the care of an ordinarily prudent person under similar circumstances, or by acts that are not in good faith, involve a conflict of interest, or otherwise constitute gross negligence.

Factual support. Mr. Sonsurkar's failure, throughout his tenure as CSO with admin access to the luxfi/* repositories, to identify and flag the LEL / LRL-PR licensing mismatch is at minimum gross negligence in the CSO duty of care. To the extent the factual record supports knowledge or willful blindness, the conduct crosses into duty-of-loyalty breach. The duty in this count runs to Satschel itself (and derivatively to its shareholders, including investors in the \$1.1B raise); Lux's claim sounds in aiding and abetting (Count V) rather than direct duty.

Remedies. Damages payable to Satschel (or its creditors / shareholders in derivative or class capacity); equitable relief including removal; punitive damages if record warrants.

Count V — Aiding and Abetting Breach of Fiduciary Duty (Delaware, against Messrs. Choi, Tromblye, Church)

Elements. (i) Existence of a fiduciary relationship; (ii) breach of fiduciary duty by the fiduciary; (iii) knowing participation in the breach by the non-fiduciary defendant; (iv) damages. *Malpiede v. Townson*, 780 A.2d 1075 (Del. 2001).

Factual support. To the extent Messrs. Choi, Tromblye, and Church had actual or constructive knowledge of the licensing mismatch and continued to direct Satschel's commercial operations notwithstanding, they aided and abetted Mr. Sonsurkar's breach.

Count VI — Unjust Enrichment

Elements. (i) Enrichment of the defendant; (ii) at plaintiff's expense; (iii) absence of justification; (iv) absence of remedy at law.

Factual support. Satschel's commercial deployment of Lux IP without an executed license enriches Satschel at Lux's expense; the LEL and LRL-PR expressly negate any justification for commercial use absent a license. Restitution should measure the value that Satschel captured, plausibly by reference to (a) the capital raised at \$1.1B attributable to the unlicensed IP, (b) the gross-revenue royalty Lux would have charged at arm's-length, or (c) the avoided build-cost Satschel saved by relying on Lux's twelve-year investment.

Count VII — Conversion

Elements. (i) Plaintiff’s interest in identifiable property; (ii) defendant’s intentional interference inconsistent with plaintiff’s rights.

Factual support. Satschel’s deployment of Lux IP without authorization, particularly the patent-protected modules under LRL-PR, constitutes intentional interference with Lux’s exclusive rights.

Count VIII — Tortious Interference with Prospective Economic Advantage

Elements. (i) Existence of a prospective economic relationship with third parties having probability of future economic benefit to plaintiff; (ii) defendant’s knowledge of the relationship; (iii) intentional and wrongful interference; (iv) actual disruption; (v) damages.

Factual support. Lux’s commercial-license pipeline with third-party ecosystem partners is impaired by Satschel’s unlicensed use of the same modules; Satschel’s offering of effectively “competitor” services on Lux IP without paying Lux’s customary license fee depresses the market price third parties will pay for their own licenses. Satschel’s white-label tenants (MLC, VCC, TZERO, ReTokens) constitute particular instances of this interference.

Count IX — Securities Fraud (Rule 10b-5)

Elements. (i) Material misrepresentation or omission; (ii) scienter; (iii) connection with the purchase or sale of a security; (iv) reliance; (v) economic loss; (vi) loss causation.

Factual support. To the extent the \$1.1B raise was supported by representations regarding Satschel’s IP ownership or licensed access that omit or mischaracterize the unlicensed status of the Lux technology, the elements are met. Satschel officers had scienter through Mr. Sonsurkar’s repository access; investors relied on the representations; the omission was material to a reasonable investor in a technology-backed venture; and loss causation is established by the value attributable to the misrepresented or omitted IP rights. Lux’s standing depends on whether Lux itself purchased Satschel securities (likely not); however, Lux may have collateral standing as a third-party beneficiary and may also pursue regulator-referral remedies. **Counsel should evaluate this count carefully — its primary utility is as collateral leverage rather than as a direct claim by Lux.**

Count X — Lanham Act False Advertising (15 U.S.C. §1125(a))

Elements. (i) False or misleading statement about defendant’s goods or services; (ii) materiality; (iii) actual deception or tendency to deceive; (iv) commerce; (v) injury.

Factual support. Satschel’s marketing materials describe its technology stack (post-quantum cryptography, FHE-based dark-pool matching, MPC custody, multi-chain architecture, Quasar-derived consensus) in terms that ascribe ownership or proprietary control to Satschel without disclosing the Lux origin or the conditional / unlicensed nature of the access. To the extent reasonable consumers are deceived as to origin or quality (or as to Satschel’s authority to provide the services), Lanham Act false-advertising claims attach.

Count XI — Common-Law Fraud

Elements. (i) False representation; (ii) knowledge of falsity or reckless disregard; (iii) intent to induce reliance; (iv) justifiable reliance; (v) injury.

Factual support. As to Lux directly: to the extent Satschel officers represented to Lux personnel (Mr. Kelling, Mr. Bin, or others) that licensing arrangements would be formalized in due course and Lux relied to its detriment by continuing to contribute engineering work without an executed agreement, common-law fraud may lie. The factual development here will depend on specific representations made to specific Lux personnel; this is a count that needs careful pleading support.

Count XII — Civil Conspiracy

Elements. (i) Two or more persons; (ii) an unlawful act; (iii) by unlawful means; (iv) an agreement; (v) damages.

Factual support. To the extent the Satschel officer team (Messrs. Choi, Tromblye, Church, and Sonsurkar) acted in concert in the commercial deployment of unlicensed Lux IP and the raising of capital on the back of that deployment, civil conspiracy lies. This count provides joint and several liability and discovery latitude.

VI. Damages Analysis

The damages owed to Lux Industries Inc. scale across three scenarios. Counsel should treat these as a framework; ranges will tighten with discovery.

Small Scenario (Path A — Negligent Posture, Settlement)

The factual record supports a Path A characterization (CSO duty-of-care failure / negligent oversight; no proven intent). Damages are the back-license amount plus a reasonable forward-looking grant:

Item	Amount
Back-license value (12 yr at \$2.5M/yr base)	\$30M
Pre-judgment interest (5%, simple)	\$10M
Royalty component (1.5% of estimated GR)	\$10M
Forward-looking license value (NPV)	\$30M
Subtotal — Small Scenario	\$80M

Medium Scenario (Path B — Mixed-Knowledge, Contested)

The factual record develops to show that one or more officers had actual knowledge of the licensing mismatch and the conduct continued unabated. Statutory multipliers apply selectively. Damages approximately 2.5–3× the Small Scenario:

Item	Amount
Small-Scenario base	\$80M
DTSA 2× exemplary on willful misappropriation	+\$30M
Patent willfulness component (selective trebling)	+\$40M
Disgorgement on \$1.1B-raise attribution (5%)	+\$55M
Attorneys' fees and costs	+\$10M
Subtotal — Medium Scenario	\$215M

Big Scenario (Path B Fully Litigated, Adverse Officer Record)

Factual record establishes intentional misappropriation by multiple officers; full statutory multipliers apply; capital-raise attribution expanded; Lanham Act and securities-fraud counts contribute:

Item	Amount
Base + interest + royalty	\$80M
DTSA full 2× exemplary	+\$80M
Patent full 3× trebling on willfulness	+\$120M
Disgorgement on \$1.1B-raise attribution (15–20%)	+\$200M
Lanham Act + Lanham fees	+\$10M
Securities fraud component (if Lux can establish standing)	+\$50M
Attorneys’ fees, costs, pre-judgment interest	+\$50M
Subtotal — Big Scenario	\$590M

Opportunity-Cost Damages: 90-Day Bandwidth Diversion

The damages framework above does not yet capture a distinct and significant category of harm: the opportunity cost to Lux Industries Inc. of having Mr. Kelling’s — and the broader Lux engineering team’s — bandwidth diverted to Satschel’s platform during a critical 90-day window. The data is unambiguous.

Quantification of contribution to Satschel (rolling 90-day window ending on the memorandum date). Audit of the liquidityio/* GitHub organization for commits attributable to Mr. Kelling (across the verified author identities zatsch, z@liquidity.io, z@satschel.com, and Kelling-signed) yields the following:

Repo	Zach commits	Total commits	Zach %
liquidityio/ats	806	1,426	57%
liquidityio/bd	388	751	52%
liquidityio/ta	208	364	57%
liquidityio/mpc	31	95	33%
liquidityio/evm	23	32	72%
liquidityio/fhe	10	14	71%
liquidityio/iam	27	29	93%
liquidityio/node	338	380	89%
liquidityio/operator	354	406	87%
liquidityio/universe	1,436	1,920	75%
liquidityio/contracts	72	176	41%
liquidityio/dex	24	32	75%
liquidityio/platform	36	51	71%
liquidityio/superadmin	243	338	72%
liquidityio/exchange	522	1,170	45%
liquidityio/console	28	38	74%
liquidityio/ui	27	143	19%
liquidityio/sdk	30	30	100%
liquidityio/sdk-go	9	9	100%
liquidityio/cli	114	117	97%
Total (90-day window)	4,726	7,521	63%

Code-churn magnitude across the largest repositories:

Repo	Files touched by Zach	Lines (added + deleted)
liquidityio/ats	64,899	~21.4M
liquidityio/bd	55,878	~5.9M
liquidityio/ta	69,321	~35.5M
liquidityio/universe	11,493	~760K
liquidityio/exchange	41,259	~3.9M
liquidityio/superadmin	24,688	~3.2M
liquidityio/node	26,755	~7.4M
liquidityio/operator	975	~71K

(Note: file counts include vendored-dependency churn; the unique- authored-line figure is materially lower than the raw line-count but of the same order of magnitude. The point is the magnitude, which is unambiguous — Mr. Kelling committed code on **55 distinct days in the ATS repository alone** during the 90-day window, i.e., six days a week of continuous output.)

Quantification of contribution to Lux upstream in the same 90-day window. Audit of the luxfi/* GitHub organization for commits attributable to Mr. Kelling in the same period yields:

Lux repo	Zach commits	Total commits (Lux team)
luxfi/node	0	179
luxfi/consensus	0	104
luxfi/crypto	0	64
luxfi/evm	0	99
luxfi/vm	0	27
luxfi/precompile	0	98
Total (90-day window)	0	571

The pattern is unambiguous. During the 90-day window in which Mr. Kelling committed 4,726 lines of work to Satschel — contributing the substantial majority of all engineering output in the company’s largest repositories — he committed **zero** to the upstream Lux repositories where he serves as Founder, Chief Cryptographer, and minority shareholder. The Lux upstream during that period was maintained by other Lux engineers without the benefit of his attention.

Value-creation rate translation. Satschel’s reported post-money valuation following its recent capital raise is approximately \$1.1 billion. To the extent the platform that supported that raise was substantially built or rebuilt by Mr. Kelling during the 90-day window analyzed here (and the contemporaneous git record establishes that it was), the implied value-creation rate of his bandwidth is approximately **\$1.1 billion / 90 days = \$12.2 million per day** of diverted attention. **The opportunity cost to Lux Industries Inc.** —measured by what Lux’s own roadmap would have produced during the same window had Mr. Kelling’s attention been available to Lux upstream — **is plausibly bounded above by the same rate, and is bounded below by the proportional share of Lux’s existing enterprise trajectory attributable to founder-level engineering input.**

Damages calculation. A reasonable measure of opportunity- cost damages is:

Component	Range
Bandwidth diversion at 50% of value-creation rate (90 days)	\$500M – \$600M
Bandwidth diversion at 25% rate	\$250M – \$300M
Bandwidth diversion at 10% rate (conservative)	\$100M – \$120M
Conservative opportunity-cost claim	\$100–300M

This opportunity-cost component is independent of, and additive to, the IP-misappropriation damages set out in the Small / Medium / Big scenarios above. The claim sounds in tortious-interference-with- contract (Lux’s implicit / fiduciary expectation of Mr. Kelling’s attention as Founder), unjust enrichment (Satschel captured value that should have flowed to Lux), and breach of fiduciary duty by Satschel officers (who knowingly directed Mr. Kelling’s bandwidth to Satschel deliverables in lieu of Lux deliverables despite his dual obligations).

This is the single most striking quantitative fact in the memorandum and should be lead-in evidence for Mr. Dupont in any opening negotiation. Satschel built itself to a \$1.1B valuation in the 90 days Mr. Kelling’s bandwidth was directed there; Lux was deprived of that same bandwidth at exactly the rate Satschel captured it.

Damages Summary (Including Opportunity Cost)

Scenario	IP damages	+ Opp. Cost
Small (Path A, settlement context)	\$80M	\$180–380M
Medium (Path B, mixed-record litigation)	\$215M	\$315–515M
Big (Path B, full-record litigation w/ multipliers)	\$590M	\$690–890M

The Big-Scenario range is supported by the Georgia-Pacific royalty framework (1.5–3.0% of gross trading + settlement revenue applied to twelve years of unlicensed use, with multipliers), the patent willfulness ceiling (3× under §284), the DTSA exemplary multiplier (2× under §1836(b)(3)(C)), and the unjust- enrichment recoupment against the recent capital raise.

VII. Trading-Fee Revenue Projections (User-Scale Analysis)

To assist Mr. Dupont’s deal modeling, the following projects Liquidity’s enterprise-value trajectory at four user-base milestones and computes Lux’s economic interest at each, under the proposed 10% equity stake (plus optional royalty under alternative structures). Three Average Revenue Per User (“ARPU”) tiers are modeled — conservative (\$50/yr), base (\$100/yr), and premium (\$200/yr) — to bracket the realistic range for a tokenized-asset exchange + ATS/BD/TA combined offering.

Annual Revenue, Implied Enterprise Value, and Lux 10% Stake Value

Scenario	Users	Revenue	Enterprise Value	Lux 10% Stake
<i>Conservative ARPU (\$50/year)</i>				
Stage 1	100K	\$5M	\$50M (10×)	\$5M
Stage 2	1M	\$50M	\$500M (10×)	\$50M
Stage 3	10M	\$500M	\$5B (10×)	\$500M
Stage 4	100M	\$5B	\$50B (10×)	\$5B
<i>Base ARPU (\$100/year)</i>				
Stage 1	100K	\$10M	\$150M (15×)	\$15M
Stage 2	1M	\$100M	\$1.5B (15×)	\$150M
Stage 3	10M	\$1B	\$15B (15×)	\$1.5B
Stage 4	100M	\$10B	\$150B (15×)	\$15B
<i>Premium ARPU (\$200/year)</i>				
Stage 1	100K	\$20M	\$400M (20×)	\$40M
Stage 2	1M	\$200M	\$4B (20×)	\$400M
Stage 3	10M	\$2B	\$40B (20×)	\$4B
Stage 4	100M	\$20B	\$400B (20×)	\$40B

Lux's 10% equity, retained through the growth curve, is worth between \$5 million (Stage 1, Conservative ARPU) and \$40 billion (Stage 4, Premium ARPU). The mid-case (Base ARPU at 100M users) implies \$15 billion of Lux equity value — which is the order of magnitude at which the user-base projection meaningfully exceeds any plausible damages award in litigation.

Trading-Fee Royalty Alternative (Approach C scenario)

Under a hybrid structure (5% equity + 5 basis-points-of-volume royalty), Lux's royalty stream at each user-base scale:

Scenario	Implied Volume	5 bps Royalty	Capped at 10% of Rev
Stage 1 (100K users)	\$5–20B	\$2.5–10M	\$0.5–2M
Stage 2 (1M users)	\$50–200B	\$25–100M	\$5–20M
Stage 3 (10M users)	\$500B–2T	\$250M–1B	\$50–200M
Stage 4 (100M users)	\$5T–20T	\$2.5B–10B	\$0.5–2B

The royalty stream provides current-period cash that the pure equity does not; at Stage 3+ it materially exceeds the equity-only NPV at similar growth assumptions.

Win-Sharing at Larger Equity Stakes

Aspirational scenarios (a 50/50 Lux-Satschel partnership, or a Lux-led majority recapitalization following a successful Big-Scenario litigation) would scale the figures above proportionately — Lux's 50% stake at Stage 4 Premium would be worth on the order of \$200 billion. These scenarios should be understood as ceiling cases for Mr. Dupont's negotiating posture, not as expected outcomes.

VIII. Strategic Partnership Upside

Lux brings substantial commercial value to a partnership-resolution of this dispute — value that is independent of the IP settlement itself, and that should be quantified and surfaced in any negotiation

as the upside Mr. Dupont can offer in exchange for favorable settlement terms.

Real-World Asset Pipeline

Lux operates the infrastructure layer behind multi-billion-dollar RWA tokenization pipelines across multiple sovereign chains. Post- SCLA, Liquidity is the natural US-regulated listing destination for these flows:

- Tokenized US securities + private equity (Carta ingestion);
- Tokenized real estate / NNN-SPV-wrapped property exposures (the `net_lease_spe` asset class);
- Tokenized commodities (gold, silver, FX, energy);
- Tokenized fund LP interests (PE / VC, hedge funds);
- Tokenized fixed-income (US Treasuries, corporate debt, sovereign debt under Reg D / Reg S);
- Tokenized music + IP royalty streams;
- Tokenized art and collectibles (\$1M+ lots).

Conservative estimate of incremental Liquidity deal flow under the SCLA: \$5–15 billion in nominal asset value over the first 12–24 months post-close.

Banking Licenses and Financial Infrastructure (via SFPB Partnership)

Lux operates a comprehensive licensed banking and financial infrastructure stack via its SF Private Bank (“SFPB”) partnership. The combined regulatory footprint covers multiple jurisdictions:

- **United States** — federal banking partner; FDIC- insured rails.
- **Canada** — chartered banking partner.
- **Singapore** — MAS-licensed (Capital Markets Services + banking) partner.
- **New Zealand** — RBNZ-licensed deposit-taker partner.
- **Sweden** — Finansinspektionen-supervised; EU passport.
- Additional jurisdictions (UK FCA, UAE VARA, Switzerland FINMA, Australia AUSTRAC) under partnership coverage.

Services Lux can deliver to Liquidity under the SCLA include Banking-as-a-Service (BaaS), Payment Service Provider (PSP) rails, and **a custody solution that materially undercuts the Fireblocks and BitGo incumbents on price** (built on the same MPC infrastructure already in Liquidity production).

Estimated independent cost for Satschel to acquire equivalent global license suite: \$57–140M in direct regulatory and capital expenditure, plus 5–7 years of timeline. This regulatory access alone exceeds the floor settlement value of any negotiating tier.

International Distribution: M7 and 1B+ End-User Reach

Through Hanzo Inc., the broader corporate family, and the Lux ecosystem partnerships, the M7 global initiatives provide distribution access to more than 1 billion end-users across consumer,

institutional, and retail channels. Post-SCLA, Lux’s international counterparty network operates as a Reg-S-compatible distribution channel for Liquidity-listed assets, with cross-network Warp messaging providing the on-chain attestation rail. This expands Liquidity’s investor base by a meaningful multiple beyond domestic US channels.

Hanzo Inc. Portfolio Precedents

Hanzo Inc., the venture studio under which the Lux Industries relationship has historically operated, has applied the IP-in-kind founder-equity model successfully to multiple spawned companies that have since reached unicorn or near-unicorn valuations:

- **Damon Motorcycles** — electric / smart-motorcycle company.
- **Triller** — short-form video / social platform.
- **Bellabeat** — women’s health wearables.
- **Unicorn Gold** — precious-metals tokenization / investment platform.
- **Lux Network (Lux Industries Inc.)** — itself a Hanzo- spawn outcome.
- **Zoo Labs Foundation / Zoo Network** — decentralized AI research network.

The model proposed for Satschel is not novel; it has produced multiple unicorn outcomes in the same corporate family. The retained- parent-equity range across the portfolio runs 10–40%, and the 10% ask in Satschel sits at the bottom of that internal range.

Ongoing IP Pipeline

Beyond the modules in production, Lux is actively developing further IP that becomes available to Satschel under any executed SCLA: AI inference + compute precompiles; advanced ZK rollup substrates; post-quantum hybrid schemes; threshold MPC advances (dynamic re-sharing); confidential-execution attestation (TEE / SEV-SNP / TDX binding to consensus). **Under a perpetual ecosystem license, Satschel receives every future Lux IP contribution as it ships, without renegotiation or additional fees.**

The \$100B–\$1T Target Institution

The combined Lux + Liquidity vision is the construction of a \$100-billion-to-\$1-trillion-class institution at the intersection of tokenized real-world assets, post-quantum-secure infrastructure, regulated US securities markets, and global distribution. The Lux deal team should treat the negotiation not as a zero-sum settlement of a past dispute, but as the founding-document of a partnership that creates value materially in excess of the negotiating range.

IX. Wait vs. Settle Analysis

The question Mr. Dupont asked is: “Is it better to wait? Liquidity’s equity becomes more valuable, and damages we are owed grow.” The legal analysis is more nuanced than the simple formulation suggests.

Forces Favoring Delay

- **Equity appreciation:** Liquidity's enterprise value likely grows; Lux's 10% appreciates with it.
- **Royalty accrual:** each year of continued unlicensed use adds approximately \$5–10M to the Small-Scenario base.
- **Capital-raise disgorgement:** future raises expand the disgorgement universe (the \$1.1B is the floor; a \$5B Series B multiplies the relevant attribution number).
- **Pattern-of-conduct evidence:** longer the conduct continues, the stronger the willfulness narrative.

Forces Favoring Settlement Now

- **Statutory caps:** DTSA multiplier caps at 2×; patent multiplier caps at 3×. These do not grow with delay.
- **Dilution risk:** every Satschel preferred round dilutes Lux's relative ownership absent strong anti-dilution protection.
- **Operational risk:** Satschel could engineer around the LEL modules over time, reducing both damages and the future-value narrative.
- **Evidence preservation:** longer the delay, greater the risk of document destruction (countered by litigation hold).
- **Tolling vs. DTSA limitations:** 3-year statute of limitations from discovery; counsel should consider whether to file a protective complaint or obtain a tolling agreement.
- **Lux's operating runway:** a \$10M+ cash settlement component funds Lux operations now; deferred value does not.
- **Negotiating leverage compounds in our favor short-term** but evaporates if Liquidity hits trouble (regulatory action, market downturn, or executive scandal at Satschel).
- **Public visibility:** Satschel's pre-IPO / Series B / exam-cycle timing makes the next 6–12 months the highest-leverage window for Lux.

Recommendation

Settle now, with strong protective terms. The optimal structure is:

1. Significant cash component (\$10M floor; higher preferable) into Lux primary investment;
2. 10% Satschel common (or higher) with weighted-average broad-based anti-dilution;
3. Pro-rata participation right on all future preferred rounds (maintains Lux's percentage through dilution);
4. Tag-along and co-sale rights (Lux participates in any liquidity event);
5. Mandatory 1+1 board composition (1 Lux director + 1 independent director added to the existing 3-person board);

6. 99-year exclusive USA-market license to Liquidity, conditional on Satschel maintaining all SCLA terms;
7. Patent grant on all current and future Lux patents covering the licensed IP.

This captures the upside of growth (through equity appreciation under anti-dilution) **without the risk of delayed enforcement**. The statutory damages multipliers do not grow with delay; the equity does. Locking in equity now with strong anti-dilution captures the future-growth narrative without the operational risks of continued reliance on a deteriorating informal arrangement.

Litigation should be held in reserve, not pursued — litigation extracts a one-time damages payment but forecloses the ongoing partnership upside. A clean settlement that includes equity participation is strictly better than a litigated damages award, *provided* the settlement terms are themselves favorable.

Counsel should not deviate from this analysis unless the diligence record develops new evidence that materially alters either the damages range (Big-Scenario evidence emerges) or the partnership-upside range (Liquidity's trajectory changes materially).

X. Settlement Triangulation (Cash / Equity / Features Matrix)

Rather than propose a single settlement, this section presents the triangulation matrix the Lux deal team can use to negotiate. The floor and ceiling reflect the damages analysis (§VI) and the partnership upside (§VIII). Mr. Dupont is in the lead on selecting the operating tier.

Settlement Sizing Tiers

Tier	Cash	Equity	Lux Nominal Value	Implied Posture
Floor (S–)	\$5M	5%	\$60M	Soft settle; preserve relationship
Small (S)	\$10M	10%	\$120M	Approach A baseline (recommended)
Medium (M)	\$20M	15%	\$185M	Path B leverage applied softly
Big (B)	\$30M	25%	\$305M	Full Path B post-investigation
Aspirational (A+)	\$50M	35%	\$435M	Litigation-credible ceiling
50/50 (max)	\$100M	50%	\$650M	Full partnership re-cap

Cash vs. Equity Mix Sensitivity

At any given nominal value, the cash/equity mix is negotiable. Cash provides current-period Lux operating runway; equity provides upside participation. The following sensitivity table fixes Lux's settlement value at the Small-Tier \$120M floor and shows the negotiating mix options:

Mix	Cash	Equity at \$1.1B
All-cash	\$120M	0%
Heavy cash	\$60M	5.5%
Balanced (recommended)	\$10M	10%
Light cash	\$5M	10.5%
All-equity	\$0	11%

The **balanced mix at \$10M cash + 10% equity is recommended** on two grounds: (a) the \$10M cash provides material Lux operating runway without overburdening Satschel's post-raise

balance sheet, and (b) the 10% equity captures the partnership upside identified in §VII. Higher cash components are achievable if Satschel prefers to deploy raised capital as settlement rather than retain it.

Feature Unlock by Settlement Tier

The level of forward-looking value Lux is willing to bring as aligned-partner scales with the settlement tier. At the Floor (S–) tier, Lux contributes only the historical IP license. At the Big (B) tier or above, Lux’s full ecosystem and infrastructure are unlocked:

Forward-value feature	S–	S	M	B	A+
LEL/LRL–PR perpetual license	✓	✓	✓	✓	✓
Patent grant on covered IP	✓	✓	✓	✓	✓
Maintenance & support of upstream OSS		✓	✓	✓	✓
99-year exclusive USA market license		✓	✓	✓	✓
1 Lux director + 1 independent seat		✓	✓	✓	✓
RWA pipeline routing into Liquidity		✓	✓	✓	✓
SFPB banking-license access (BaaS, PSP)			✓	✓	✓
Custody (Fireblocks-undercutting)			✓	✓	✓
International M7 distribution channels				✓	✓
Closed-source (luxcpp) GPU access					✓
Co-marketing + brand alignment				✓	✓
Joint go-to-market programs				✓	✓

Mr. Dupont can use this matrix in negotiation directly: “Lux is willing to bring [feature set] at [tier], or alternatively to bring [smaller feature set] at [smaller tier].” Satschel chooses its tier; Lux’s deliverables scale accordingly.

Damages-vs-Settlement Triangulation

Outcome path	Lux receives	Satschel pays
Settlement at Floor (S–)	\$60M nominal	\$5M cash + 5% equity
Settlement at Small (S, recommended)	\$120M nominal	\$10M cash + 10% equity
Settlement at Medium (M)	\$185M nominal	\$20M cash + 15% equity
Settlement at Big (B)	\$305M nominal	\$30M cash + 25% equity
Settlement at Aspirational (A+)	\$435M nominal	\$50M cash + 35% equity
Litigation: Small-Scenario damages (Path A)	\$80M damages	\$80M cash, no equity
Litigation: Medium-Scenario damages (Path B mixed)	\$215M damages	\$215M, no equity
Litigation: Big-Scenario damages (Path B full)	\$590M damages	\$590M, no equity

Key insight: a settled outcome at the Small tier (\$120M nominal) is materially less expensive to Satschel than the Small- Scenario litigated damages (\$80M cash), *provided the equity component is properly valued*. Litigation extracts cash without preserving the partnership relationship and without giving Satschel the forward-looking benefits in §VIII. **Settlement dominates litigation at every tier for both sides.**

XI. Mandatory Terms (Lux's Floor in Any Settlement)

The following terms are non-negotiable. Any settlement at any tier must include each of them. Mr. Dupont and Mr. Donohue should treat these as the bottom of Lux's negotiating range; any Satschel pushback on them is a walk-away signal.

1. Board composition — 1 Lux director + 1 independent director added to the existing 3-person Satschel board. The Lux director seat is for Lux Industries Inc.’s direct representative (Mr. Dupont’s nominee or Mr. Pahlavi’s designate). The independent director seat is a mutually agreed nominee with no prior relationship to either side, serving as the ongoing governance anchor for the SCLA terms. Effective on closing. No ownership-maintenance condition attaches to either seat.

2. 99-year exclusive USA-market license to Liquidity for the LEL/LRL-PR portfolio in scope. Conditional on Satschel maintaining all SCLA covenants (royalty payments, anti-dilution, governance terms, reciprocity grants). Termination of the SCLA for material breach by Satschel terminates the exclusivity (license reverts to non-exclusive). Outside the USA, Lux retains the right to license the same IP to other parties.

3. Anti-dilution: weighted-average broad-based, with full ratchet on any down-round below the Lux issuance price.

4. Pro-rata participation right on all future Satschel preferred financings.

5. Tag-along and co-sale rights on any founder transfer exceeding 1% of fully-diluted equity.

6. Information rights: quarterly unaudited financials; annual audited financials; copies of all board materials, minutes, and consent resolutions; copies of all share-issuance and option- grant resolutions; access to counsel and accountants under standard NDA.

7. Patent grant on all current and future Lux patents covering the licensed IP. Worldwide; perpetual; royalty-free under the SCLA; expressly inclusive of any Lux patents not yet filed at SCLA execution.

8. Maintenance and support of upstream Lux OSS: continuing on the same shared-engineering basis as today, with no separate maintenance fee.

9. Reciprocity: Lux-network L1/L2/L3/L4 chains that maintain Lux consensus and Warp interop receive free non-exclusive license to the Liquidity protocol (ATS/BD/TA). Lux chains may offer Liquidity- style securities products without separate license.

10. Engineering commitment from Satschel: activate the Lux Warp precompile (canonical address 0x020000000000000000000000000000000005) on every Liquid chain at the next chain-upgrade window. New Liquid chains include the `warpConfig` block in genesis.

11. Litigation hold and document preservation: Satschel agrees to preserve all records, communications, and code review artifacts relating to the licensing posture from 2014 to the SCLA execution date.

12. Independent investigation cooperation: Satschel cooperates fully with any independent investigation of officer conduct undertaken by the Satschel disinterested directors or their appointed counsel.

XII. Special Note Re: Mr. Zach Kelling

Mr. Kelling occupies a position requiring particular care in Lux’s litigation and settlement strategy. The factual context established in §IV(F) bears directly here: Mr. Kelling was the *remediator* of the pre-Kelling Satschel technical and security failures (the broken time-price priority matching; the \$40,000 ACH- exploit theft) and the architect of the rebuilt platform that subsequently supported the \$1.1B raise. He was not the author of the conduct giving rise to this matter; he is, if anything, the party whose work was misappropriated alongside Lux’s. The further facts:

- Mr. Kelling founded both Lux Industries Inc. and Satschel, Inc., and authored substantial portions of the Lux IP at issue.
- He is currently the Architect and Chief Cryptographer of Satschel, but *not* its CEO or CTO. He previously held those titles in earlier corporate periods.
- He is a **minority shareholder of Lux Industries Inc.**
- He holds **zero equity in Satschel / Liquidity.io.**
- He is not a director on either side.
- He has been operationally displaced from Satschel’s commercial direction; the operational conduct giving rise to this matter is attributable to other officers (Messrs. Choi, Tromblye, Church, and Sonsurkar) and not to him.

Litigation-strategy implications:

- Mr. Kelling should *not* be named as a defendant in any Lux suit. The conduct alleged here is not his.
- Mr. Kelling is likely a material witness for Lux as to (a) the authorship and ownership of the Lux IP, (b) the absence of any executed commercial license between Lux and Satschel, and (c) the operational displacement narrative explaining why the licensing mismatch was not raised internally on his initiative.
- Counsel should consider whether a litigation-cooperation agreement, witness-protection arrangement, or formal indemnification of Mr. Kelling (as a non-defendant witness) is appropriate to ensure his continued availability as a witness without undue personal exposure.
- In any public communication regarding the matter, Lux should take care not to attribute the misconduct to Mr. Kelling. The narrative should be specific to the Satschel officers actually responsible.

- In the SCLA itself, Lux should consider providing for Mr. Kelling’s protective treatment, including (a) a release of any claims Lux might assert against him in his personal capacity, (b) acknowledgement of his historical authorship contribution, and (c) a ROFR or similar mechanism if his Satschel architectural role generates IP that should flow back to Lux upstream.

Practical effect: Mr. Kelling can serve as a productive intermediary in the Lux–Satschel negotiation, given his founder status at both entities, his historical relationships, and his absence of financial interest in Satschel that would compromise his neutrality. Mr. Dupont should consider engaging Mr. Kelling directly in informational meetings during the negotiation phase, subject to outside counsel’s oversight.

XIII. Procedural Recommendations

Decision chain: Lux’s decision flow on this matter is:

1. **Mr. Pahlavi (President & CEO)** — final approval authority on any executed SCLA, any litigation filing, and any strategic public communication.
2. **Mr. Dupont (Managing Partner, Strategic Transactions; largest investor)** — deal lead. Conducts the negotiation, selects the settlement tier, and presents the executed term sheet to Mr. Pahlavi.
3. **Mr. Donohue (General Counsel)** — legal lead. Drafts the SCLA, supervises outside counsel, manages litigation hold and document preservation, leads any pre-suit investigation, and prepares the SCLA for execution by Mr. Pahlavi.
4. **Mr. Bin (CTO; this memorandum’s author)** — operational and technical support. Provides factual development on the IP portfolio, the licensing record, and the Satschel-side operational context.

Recommended outside engagement:

- **Outside litigation counsel:** engage a firm with demonstrated DTSA + patent + Delaware-fiduciary-duty experience. Recommended: Wilson Sonsini, Cooley, Fenwick & West, or Quinn Emanuel (for the litigation posture).
- **Valuation firm:** engage an independent firm to opine on the fairness of any equity-for-IP allocation in the SCLA. Recommended: Houlihan Lokey, Duff & Phelps (Kroll), or a specialist boutique with digital-asset IP experience.
- **Investigative firm:** engage a separate firm to investigate Satschel officer conduct during the relevant period (repository access, code review, internal communications, license discussions). The investigation report informs both Lux’s settlement posture and any subsequent litigation. Recommended: Kroll Investigations, FTI Consulting, or specialist tech-IP investigators.

Recommended timeline:

Day	Action
0	Mr. Pahlavi approves the memorandum and authorizes the deal team to proceed.
3	Outside counsel engaged.
7	Litigation hold notice transmitted to Satschel and named officers.
14	Investigation firm begins document review.
21	Term sheet delivered to Satschel for the recommended settlement tier.
30	Satschel disinterested-director response received.
45	Investigation report finalized; settlement terms tightened or expanded based on findings.
60	Definitive SCLA executed.
75	Closing: cash payment, share issuance, license activation.

If the timeline slips materially, the deal team should escalate to Mr. Pahlavi for guidance on whether to (a) extend negotiation, (b) pivot to Path B leverage (a more aggressive litigation-credible posture), or (c) file a protective complaint to preserve statutory rights while continuing negotiation.

XIV. Prayer for Relief (Template)

In the event Lux Industries Inc. proceeds to litigation, the prayer for relief in the complaint will request:

1. Compensatory damages on each Count, in an amount to be proven at trial, anticipated in the Small-to-Big Scenario range of \$80M to \$590M.
2. Disgorgement of unjust enrichment derived from Satschel's \$1.1 billion capital raise, in an amount attributable to the misappropriated Lux IP.
3. Reasonable royalty under 35 U.S.C. §284 (patent) and 18 U.S.C. §1836(b)(3)(B) (DTSA).
4. Exemplary damages of 2× on the DTSA count (18 U.S.C. §1836(b)(3)(C)) for willful and malicious misappropriation.
5. Treble damages on the patent count (35 U.S.C. §284) for willful infringement.
6. Statutory damages on the copyright count (17 U.S.C. §504(c)) at the elected maximum.
7. Attorneys' fees and costs under 18 U.S.C. §1836(b)(3)(D), 17 U.S.C. §505, 35 U.S.C. §285, and 15 U.S.C. §1117(a).
8. Preliminary and permanent injunctive relief enjoining Satschel from further use of the misappropriated IP absent an executed commercial license.
9. Pre-judgment and post-judgment interest.
10. Such other and further relief as the Court deems just and proper.

Demand for jury trial on all issues so triable.

SCHEDULES

Schedule A — Lux IP Inventory in Production at Satschel

The full inventory is maintained at `~/work/lux/legal/LIQUIDITY_LICENSE_AUDIT_2026_05_25.md`.
Headline:

- BSD-3/Apache-2.0/MIT modules (no Lux action): approximately 25 modules.
- LEL modules (commercial license required absent Authorized- Network qualification): `aml`, `broker`, `captable`, `compliance`, `corona`, `crypto`, `fhe`, `hsm`, `maker`, `mpc`, `precompile` (and sub-precompiles), `ringtail`, `threshold`, `transfer`, `treasury`, `warp`.
- LRL-PR modules (research-only by default; commercial license required): `vm`, `staking`, `sampler`, `dex` (Go bindings).
- LGPL modules (compliance obligations independent of commercial license): `evm`.
- Modules without a LICENSE file (default copyright; no rights granted absent affirmative license): `cex` and a small number of related modules; counsel should treat as no-rights-granted pending upstream remediation.

Schedule B — Lux IP Pipeline (Not Yet in Production at Satschel, Available Under SCLA)

- AI inference and compute precompiles (LP-9080 family).
- Advanced ZK rollup substrates.
- Post-quantum hybrid signature schemes (X-Wing + ML-KEM; hybrid ML-DSA + ECDSA).
- Threshold MPC dynamic re-sharing protocols.
- Confidential-execution attestation (TEE / SEV-SNP / TDX binding to consensus).
- White-label partner portal infrastructure (currently powering MLC, VCC, TZERO, ReTokens).
- Closed `~/work/luxcpp/*` (CUDA / Metal / WebGPU / FPGA GPU accelerators, C++ DEX matching engine, AI compute kernels) — available under separate Private-Moat Addendum.

Schedule C — Banking License Cost-to-Replicate (SFPB Equivalent)

License	Estimated Cost
US national or state banking charter	\$20–50M
Canadian Schedule II / partnership	\$10–20M
MAS Capital Markets Services + banking (Singapore)	\$5–10M
RBNZ deposit-taker (New Zealand)	\$5–15M
Sweden Finansinspektionen (EU passport)	\$5–10M
UK FCA EMI / Payment Institution	\$2–5M
Switzerland FINMA digital-asset banking	\$5–15M
Hong Kong SFC Type 1 / Type 7 + VATP	\$5–15M
Total direct regulatory + capital	\$57–140M
Plus 5–7 years of timeline + execution risk	

Schedule D — Hanzo Inc. Portfolio Precedents

- Damon Motorcycles — electric / smart-motorcycle company.
- Triller — short-form video / social platform.
- Bellabeat — women’s health wearables and AI-driven personal health platform.
- Unicorn Gold — precious-metals tokenization / investment platform.
- Lux Network (Lux Industries Inc.) — subject of this memorandum.
- Zoo Labs Foundation / Zoo Network — decentralized AI research network.

Retained-parent-equity range across the portfolio: 10%–40% in spawned operating companies. The 10% ask in Satschel is at the bottom of that internal range.

Schedule E — Mandatory Terms (Lux’s Floor)

See §XI, above.

End of memorandum. This document is confidential, privileged, and prepared in anticipation of litigation as well as for board strategy purposes. The proposed settlement framework contained herein is non-binding until executed as a definitive Settlement and Cross-License Agreement. Distribution outside Lux Industries Inc., its Board of Directors, the named decision-chain officers (Messrs. Pahlavi, Dupont, Donohue, and Bin), and engaged outside counsel requires the express written authorization of Mr. Pahlavi and Mr. Donohue.